Assignment Unit :- 1

First order ordinary differential equations

1) Solve $(2xy^4 + \sin y)\,dx + (4x^2y^3 + x.\cos y).\,dy = 0$

→ $M\,dx + N\,dy = 0$

$\therefore M = 2xy^4 + \sin y$

$\therefore N = 4x^2y^3 + x.\cos y$

$$\frac{\partial M}{\partial y} = 8xy^3 + \cos y \quad \text{—— (1)}$$

$$\frac{\partial N}{\partial x} = 8xy^3 + \cos y \quad \text{— (2)}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$ —— ∴ This is exact equation

∴ General Solution is

$\int M.dx + \int N.dy = c$

$\int 2xy^4 + \sin y\,.dx + \int 0\,.dy = c$

$\cancel{2}\frac{x^2}{\cancel{2}}y^4 + x.\sin y = c$

$$\boxed{x^2y^4 + x.\sin y = c}$$

2) Solve $(3y^2 + 10xy^2) \cdot dx + (6xy - 2 + 10x^2y) \cdot dy = 0$

Sol^n:

$$M dx + N dy = 0$$

$$\therefore M = 3y^2 + 10xy^2$$

$$\therefore N = 6xy - 2 + 10x^2y$$

$$\frac{\partial M}{\partial y} = 6y + 20xy \quad \text{—— (1)}$$

$$\frac{\partial N}{\partial x} = 6y + 20xy \quad \text{—— (2)}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$ This is exact eq^n

$\therefore$ General Solution is.

$$\int M \cdot dx + \int N \cdot dy = C$$

$$\int 3y^2 + 10xy^2 \cdot dx + \int -2 \, dy = C$$

$$3y^2 \int 1 \, dx + 10y^2 \int x \cdot dx - 2 \int 1 \cdot dy = C$$

$$3y^2 x + 10y^2 \frac{x^2}{2} - 2y = C$$

$$\boxed{3y^2x + 5y^2x^2 - 2y = C}$$

3) Solve $(y\sec^2 x + \sec x \tan x)\cdot dx + (\tan x + 2y)\cdot dy = 0$

Sol →

$M\,dx + N\,dy = 0$

$\therefore M = y\cdot \sec^2 x + \sec x \cdot \tan x$

$\therefore N = \tan x + 2y.$

$$\frac{\partial M}{\partial y} = \sec^2 x \quad \text{—} \quad (1)$$

$$\frac{\partial N}{\partial x} = \sec^2 x \quad \text{—} \quad (2)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This is exact equation.

$\therefore$ General solution is

$$\int M\cdot dx + \int N\cdot dy = c$$

$$\int y\cdot \sec^2 x + \sec x\cdot \tan x\, dx + \int 2y\cdot dy = c$$

$$y\int \sec^2 x + \int \sec x\cdot \tan x\, dx + 2y\int y\, dy = c$$

$$y\tan^2 x + \sec x + \cancel{2yx} \cancel{=c} \; 2\,\frac{y^2}{2} = c$$

$$\boxed{y\tan^2 x + \sec x + y^2 = c}$$

4) Solve $(x^2y - 2xy^2)\,dx + (3x^2y - x^3)\,dy = 0$ → (1)

→ $M = x^2y - 2xy^2$

$N = 3x^2y - x^3$

$\frac{\partial x}{\partial y} = \cancel{x^2y - 2}$

$\frac{\partial M}{\partial y} = x^2 - 4xy$

$\frac{\partial N}{\partial y} = 6xy - 3x^2$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ This is not exact equation.

∵ Now, we have to apply homogenous rule as we know.

$$Mx + Ny \neq 0$$

$$I.f. = \frac{1}{(x^2y - 2xy^2)x + (3x^2y - x^3)y}$$

$$\cancel{x^3y - 2x^2y^2 + 3x^2y^2 - x^3y}$$

$$I.f \quad \frac{1}{x^2y^2}$$

Multiplying I.f in eqn (1)

$$\left(\frac{x^2y - 2xy^2}{x^2y^2}\right)dx + \left(\frac{3x^2y - x^3}{x^2y^2}\right)dy = 0$$

$$\left(\frac{x^2y}{x^2y^2} - \frac{2xy^2}{x^2y^2}\right)dx + \left(\frac{3x^2y}{x^2y^2} - \frac{x^3}{x^2y^2}\right)dy = 0$$

$$\left(\frac{1}{y} - \frac{2}{x}\right)dx + \left(\frac{3}{y} - \frac{x}{y^2}\right)dy = 0$$

This is exact equation

$\therefore$ General solution is

$$\int\left(\frac{1}{y}-\frac{2}{x}\right)dx + \int\left(\frac{3}{y}\right)\cdot dy = c$$

$$\int\frac{1}{y}\cdot dx - \int\frac{2}{x}\cdot dx + \int\frac{3}{y}\cdot dy = c$$

$$\frac{1}{y}\int 1\cdot dx - 2\int\frac{1}{x}\cdot dx + 3\int\frac{1}{y}\cdot dy = c$$

$$\frac{1}{y}x - 2\log x + 3\log y = c$$

$$\frac{x}{y} - \log x^2 + \log y^3 = c$$

$$\frac{x}{y} - \log\left(\frac{y^3}{x^2}\right) = c$$

5) Solve $(xy+y^2)\,dx - x^2\,dy = 0$

$$(xy+y^2)\,dx - (x^2)dy = 0 \longrightarrow (1)$$

$M = xy+y^2$

$N = -x^2$

$$\frac{\partial M}{\partial y} = x+2y$$

$$\frac{\partial N}{\partial x} = -2x$$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ This is not ~~equal~~ ~~to~~ exact equation.

to convert it into exact form we have to multiply eq^n ① by I.F.

$$I.F = \frac{1}{Mx+Ny} = \frac{1}{(xy+y^2)x - (x^2)y}$$

$$\frac{1}{x^2y+y^2x - x^2y}$$

$$I.f. = \frac{1}{y^2x}$$

Multiply I.f in eq^n ①

$$\cancel{\frac{1}{xy+y^2}}$$

$$\left(\frac{xy+y^2}{y^2x}\right)dx - \left(\frac{x^2}{y^2x}\right)dy = 0$$

$$\left(\frac{xy}{y^2x} + \frac{y^2}{y^2x}\right)dx - \left(\frac{x^2}{y^2x}\right)dy = 0$$

$$\left(\frac{1}{y} + \frac{1}{x}\right)dx - \left(\frac{x}{y^2}\right)dy = 0 \quad ②$$

Now this eqn ② is exact eqn.

we can found G.S

$$\int\left(\frac{1}{y}+\frac{1}{x}\right)dx-\int\left(\frac{x}{y}\right)dy=c$$

$$\frac{1}{y}\int 1\cdot dx+\int\frac{1}{x}\,dx-\int 0\,dy=c$$

$$\boxed{\frac{x}{y}+\log x=c}$$

6) Solve $(x^2y^2+5xy+2)y\cdot dx+(x^2y^2+4xy+2)x\cdot dy=0$.

Sol^n:- $(x^2y^2+5xy+2)y\cdot dx+(x^2y^2+4xy+2)x\cdot dy=0$. ------ ①

$$M=(x^2y^2+5xy+2)y = x^2y^3+5xy^2+2y$$
$$N=(x^2y^2+4xy+2)x = x^3y^2+4x^2y+2x$$

$$\frac{\partial M}{\partial y}=3x^2y^2+10xy+2$$

$$\frac{\partial N}{\partial x}=3x^2y^2+8xy+2$$

$\frac{\partial M}{\partial y}\neq\frac{\partial N}{\partial x}$ This is not exact eqn.

~~Multiply I.f in eqn ①~~

To Convert D.E eqn ① into exact D.E.

$$I.F = \frac{1}{Mx - Ny} = \frac{1}{(x^2y^2+5xy+2)y\,x - (x^2y^2+4xy+2)xy}$$

$$= \frac{1}{\cancel{x^3y^3}+8x^2y^2+\cancel{2xy}-\cancel{x^3y^3}-4x^2y^2-\cancel{2xy}}$$

$$I.f = \frac{1}{x^2y^2}$$

Multiply I.f in eqn ①

$$\int\left(\frac{(x^2y^2+5xy+2)y}{x^2y^2}\right)dx + \left(\frac{(x^2y^2+4xy+2)x}{x^2y^2}\right)dy = c$$

$$\left(\frac{x^2y^3+5xy^2+2y}{x^2y^2}\right)dx + \left(\frac{x^3y^2+4x^2y+2x}{x^2y^2}\right)dy = c$$

$$\left(\frac{\cancel{x^2}y^{\cancel{3}}}{\cancel{x^2}y^{\cancel{2}}} + \frac{5\cancel{xy^2}}{\cancel{x^2y^2}} + \frac{2\cancel{y}}{x^2y^{\cancel{2}}}\right)dx + \left(\frac{x^{\cancel{3}}y^{\cancel{2}}}{\cancel{x^2y^2}} + \frac{4\cancel{x^2}y}{\cancel{x^2}y^2} + \frac{2\cancel{x}}{x^{\cancel{2}}y^2}\right)dy = c$$

$$\left(y + \frac{5}{x} + \frac{2}{x^2y}\right)dx + \left(x + \frac{4}{y} + \frac{2}{xy^2}\right)dy = c$$

$$\int\left(y + \frac{5}{x} + \frac{2}{x^2y}\right)dx + \int \frac{4}{y}\cdot dy = c$$

$$\int y\,dx + \int \frac{5}{x}\,dx + \int \frac{2}{x^2y}\,dx + \int \frac{4}{y}\cdot dy = c$$

$$y\int 1\cdot dx + 5\int \frac{1}{x}\,dx + \frac{2}{y}\int \frac{1}{x^2}\,dx + 4\int \frac{1}{y}\cdot dy = c$$

$$yx + 5\log x + \frac{2}{y} - \frac{1}{x} + 4\log y = c$$

$$yx + \log x^5 + \frac{2}{y} - \frac{1}{x} + \log y^4 = c$$

$$\boxed{yx + \log\left(\frac{x^5}{y^4}\right) - \frac{2}{yx} = c}$$

7) Solve $(x^2y^2+2)y \cdot dx + (2-2x^2y^2)x \cdot dy = 0$

Sol^n: $(x^2y^2+2)y \cdot dx + (2-2x^2y^2)x \cdot dy = 0$ ..... (1)

$$M = (x^2y^2+2)y$$

$$N = (2-2x^2y^2)x$$

$$\frac{\partial M}{\partial y} = 3x^2y^2 + 2$$

$$\frac{\partial N}{\partial x} = 2 - 6x^2y^2$$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ This is not exact eq^n

$$\cancel{I.f = \frac{1}{(x^2y^2+2)xy - (2-2x^2y^2)xy}}$$

$$I.f = \frac{1}{x^3y^3 + \cancel{2xy} - \cancel{2xy} + 2x^3y^3}$$

$$\boxed{I.f = \frac{1}{3x^3y^3}}$$

Multiply I.f in eqn ①

$$\cancel{x^2y^2+5x}$$

$$\frac{(x^2y^2+2)x}{3x^3y^2}\,dx + \frac{(2-2x^2y^2)x}{3x^2y^3}\,dy = 0.$$

$$\left(\frac{x^2y^2}{3x^3y^2} + \frac{2}{3x^3y^2}\right)dx + \left(\frac{2}{3x^2y^3} - \frac{2x^2y^2}{3x^2y^3}\right)dy = 0$$

$$\left(\frac{1}{3x} + \frac{2}{3x^3y^2}\right)dx + \left(\frac{2}{3x^2y^3} - \frac{2}{3y}\right)dy = 0$$

above eqn is exact

Now, find G.S.

$$\int\left(\frac{1}{3x} + \frac{2}{3x^3y^2}\right)dx + \int -\frac{2}{3y}\,dy = c$$

$$\frac{1}{3}\int\frac{1}{x}dx + \frac{2}{3y^2}\int\frac{1}{x^3}\,dx - \frac{2}{3}\int\frac{1}{y}\,dy = c$$

$$\frac{1}{3}\log x + \frac{2}{3y^2}\left(-\frac{1}{2x^2}\right) - \frac{2}{3}\log y = c$$

$$\frac{\log x}{3} - \frac{1}{3y^2x^2} - \frac{2\log y}{3} = c$$

<u>multiply by 3</u>

$$\log x - \frac{1}{y^2x^2} - 2\log y = c$$

$$\boxed{\log\left(\frac{x}{y^2}\right) - \frac{1}{y^2x^2} = c}$$

Q8 Solve $(x^2+y^2+1).dx - (2xy)\,dy = 0$

Sol^n:- given :- $(x^2+y^2+1)\,dx - (2xy)\,dy = 0 \longrightarrow (1)$

$M = x^2+y^2+1$
$N = -2xy$

$$\frac{\partial M}{\partial y} = 2y$$

$$\frac{\partial N}{\partial x} = -2y$$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ This is not exact equation.

Now;

$$\left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}\right) = \frac{2y-(-2y)}{-2xy}$$

$$= \frac{4y}{-2xy}$$

$$\left(\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}\right) = -\frac{2}{x} = f(x)$$

$$\therefore \; I.F = e^{\int f(x)\,dx}$$
$$= e^{\int -\frac{2}{x}\cdot dx}$$
$$= e^{-2\int \frac{1}{x}\cdot dx}$$

$= e^{-2\log x}$

$= e^{\log x^{-2}}$

$I.F = x^{-2}$

$I.F = \frac{1}{x^2}$

Multiply I.F in eqn ①

$$\frac{1}{x^2}\left[(x^2+y^2+1)\,dx - (2xy)\,dy\right] = 0$$

$$\left(\frac{x^2+y^2+1}{x^2}\right)dx - \left(\frac{2xy}{x^2}\right)dy = 0$$

$$\left(\frac{x^2}{x^2} + \frac{y^2}{x^2} + \frac{1}{x^2}\right)dx - \left(\frac{2y}{x}\right)dy = 0$$

$$1 + \frac{y^2}{x^2} + \frac{1}{x^2}\cdot dx - \frac{2y}{x}\cdot dy = 0 \longrightarrow ②$$

eqn ② is exact eqn.

$\therefore M^* dx \,.\, N^* dy = c$

$$\int 1 + \frac{y^2}{x^2} + \frac{1}{x^2}\cdot dx - \int 0 \cdot dy = c$$

$$\int 1\cdot dx + y^2\int \frac{1}{x^2}dx + \int \frac{1}{x^2}\cdot dx = c$$

$$x + y^2\left(-\frac{1}{x}\right) - \frac{1}{x} = c$$

$$x - \frac{y^2}{x} - \frac{1}{x} = c$$

$$\frac{x^2-y^2-1}{x} = c \qquad \boxed{x^2-y^2-1 = cx}.$$

Q9) Solve $(x^2+y^2+x)\,dx + xy \cdot dy = 0$

→ Given $(x^2+y^2+x)\,dx + xy \cdot dy = 0$ ---- (1)

$M = x^2+y^2+x$ $\qquad$ $N = xy$

$\frac{\partial M}{\partial y} = 2y$ $\qquad$ $\frac{\partial N}{\partial x} = y$

Now, find f(x)

$$f(x) = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$$

$$f(x) = \frac{2y - y}{xy}$$

$$f(x) = \frac{\cancel{y}}{x\cancel{y}}$$

$$\boxed{f(x) = \frac{1}{x}}$$

~~Now, I.f is~~

Now, find I.f.

$$I.f = e^{\int f(x)\,dx}$$
$$= e^{\int \frac{1}{x}\,dx}$$
$$= e^{\log x}$$

$$\boxed{I.f = x}$$

multiply I.f in eq^n ①

$$x(x^2+y^2+x)\,dx + x(xy)\,dy = 0$$

$$(x^2+y^2x+x^2)\,dx + x^2y\,dy = 0.$$

Now, sol^n D.E

$$\int (x^3+xy^2+x^2)\,dx + \int 0\cdot dy = c$$

$$\int x^3dx + \int xy^2dx + \int x^2\,dx = c$$

$$\frac{x^4}{4} + \frac{x^2y^2}{2} + \frac{x^3}{3} = c$$

multiply by 12

$$\cancel{12}^{3}\frac{x^4}{\cancel{4}} + \cancel{12}^{6}\frac{x^2y^2}{\cancel{2}} + \cancel{12}^{4}\frac{x^3}{\cancel{3}} = c$$

$$\boxed{3x^4 + 6x^2y^2 + 4x^3 = c}$$

Q10) Solve $(2x^2y^2+ye^x)\,dx + (-e^x-y^3)\,dy = 0$

→ Given :- $(2x^2y^2+ye^x)\,dx + (-e^x-y^3)\,dy = 0$ .... ①

$M = 2x^2y^2 + ye^x$ $\qquad$ $N = -e^x - y^3$

$\frac{\partial M}{\partial y} = 4x^2y + e^x$ $\qquad$ $\frac{\partial N}{\partial x} = -e^x$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

eq^n is not exact D.E.

$$f(x) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{-e^x - 4x^2y - e^x}{2x^2y^2 + ye^x}$$

$$= \frac{-2e^x - 4x^2y}{2x^2y^2 + ye^x}$$

$$= \frac{-2(e^x - 2x^2y)}{y(2x^2y + e^x)}$$

$$f(x) = \frac{-2}{y}$$

$$I.f = e^{\int f(y)\,dy}$$

$$= e^{\int -\frac{2}{y}\,dy}$$

$$= e^{-2\int \frac{1}{y}\,dy}$$

$$= e^{-2\log y}$$

$$= e^{\log y^{-2}}$$

$$= y^{-2}$$

$$I.f = \frac{1}{y^2}$$

multiply I.f to the eqn (I)

$$\left(\frac{2x^2y^2 + ye^x}{y^2}\right)dx + \left(\frac{-e^x - y^3}{y^2}\right)dy = 0$$

$$\frac{2x^2y^2}{y^2} + \frac{ye^x}{y^2}\,dx + \frac{-e^x}{y^2} - \frac{y^3}{y^2}\,dy = 0$$

$$2x^2 + \frac{e^x}{y}\,dx + \frac{-e^x}{y^2} - y \cdot dy = 0$$

above exact eqn.

Now,

$$\int_{y=\text{const}} \left(2x^2 + \frac{e^x}{y}\right) dx + \int -y \, dy = c$$

$$2\int x^2 + \frac{1}{y}\int e^x + \int -y \, dy = c$$

$$2\,\frac{x^3}{3} + \frac{1}{y}e^x + \left(-\frac{y^2}{3}\right) = c$$

$$\boxed{\frac{2x^3}{3} + \frac{e^x}{y} - \frac{y^2}{3} = c}$$

Q11) Solve $(y^4+2y)\,dx + (xy^3 + 2y^4 - 4x)\,dy = 0$

→ Given $(y^4+2y)\,dx + (xy^3+2y^4-4x)\,dy = 0$ .... ①

$M = y^4+2y$ $\qquad$ $N = xy^3+2y^4-4x$

$\frac{\partial M}{\partial y} = 4y^3+2$ $\qquad$ $\frac{\partial N}{\partial x} = y^3-4$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$ given eqn is not exact D.E.

$$\therefore f(x) = \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = \frac{y^3-4-(4y^3+2)}{y^4+2y}$$

$$= \frac{y^3-4-4y^3-2}{y^4+2y}$$

$$= \frac{-6-3y^3}{y^4+2y}$$

$$= \frac{-6-3}{3y}$$

$$= \frac{-9}{3y}$$

$$\boxed{f(x) = -\frac{3}{y}}$$

$$I.f = e^{\int f(y)\,dy}$$
$$= e^{\int -\frac{3}{y}\,dy}$$
$$= e^{-3\int \frac{1}{y}\,dy}$$
$$= e^{-3\log y}$$
$$= e^{\log y^{-3}}$$
$$= y^{-3}$$
$$\boxed{I.f = \frac{1}{y^3}}$$

Multiply I.f in eqn ①

$$\frac{(y^4+2y)}{y^3}\,dx + \frac{(xy^3+2y^4+4x)}{y^3}\,dy = 0$$

$$\frac{y^4}{y^3} + \frac{2y}{y^2}\,dx + \frac{xy^3}{y^3} + \frac{2y^4}{y^3} + \frac{4x}{y^3}\,dy = 0$$

$$\left(y + \frac{2}{y^2}\right)dx + \left(x + 2y + \frac{4x}{y^3}\right)dy = 0$$

now:

$$\int_{y=\text{const}} y + \frac{2}{y^2}\,dx + \int 2y\,dy = c$$

$$y\int 1.dx + \frac{2}{y^2}\int 1.\,dx + 2\int y\,dy = c$$

$$xy + \frac{2x}{y^2} + \cancel{2}\,\frac{y^2}{\cancel{2}} = c$$

$$\boxed{xy + \frac{2x}{y^2} + y^2 = c}$$

Q12) Solve $(y^3 - 2x^2y)\,dx + (2xy^2 - x^3)\,dy = 0$

→ Given $(y^3 - 2x^2y)\,dx + (2xy^2 - x^3)\,dy = 0$ ---- ①

$M = y^3 - 2x^2y$ $\qquad$ $N = 2xy^2 - x^3$

$\frac{\partial M}{\partial y} = 3y^2 - 2x^2$ $\qquad$ $\frac{\partial N}{\partial x} = 2y^2 - 3x^2$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$I.f = \frac{1}{Mx + Ny} = \frac{1}{(y^3 - 2x^2y)x + (2xy^2 - x^3)y}$$

$$= \cancel{\frac{1}{xy^3 - 2x^2y + 2xy^3 - x^3y}}$$

$$= \frac{1}{xy^3 - 2x^2y + 2xy^3 - x^3y}$$

$$= \frac{1}{3xy^3 - 3x^3y}$$

$$\boxed{I.f = \frac{1}{3xy(y^2 - x^2)}}$$

multiply I.f to the eqⁿ ①

$$\left(\frac{y^3 - 2x^2y}{3xy(y^2 - x^2)}\right)dx + \left(\frac{2xy^2 - x^3}{3xy(y^2 - x^2)}\right)dy = 0$$

multiply by 3

$$\left(\frac{y(y^2 - 2x^2)}{xy(y^2 - x^2)}\right)dx + \frac{x(2y^2 - x^2)}{xy(y^2 - x^2)}\,dy = 0$$

$$\frac{y^2 - 2x^2}{x(y^2 - x^2)}\,dx + \frac{2y^2 - x^2}{y(y^2 - x^2)}\,dy = 0$$

$$\frac{(y^2-x^2)-x^2}{x(y^2-x^2)}\,dx + \frac{(y^2-x^2)+y^2}{y(y^2-x^2)}\,dy = 0$$

$$\frac{\cancel{y^2-x^2}}{x\cancel{(y^2-x^2)}} - \frac{x^{\cancel{2}}}{\cancel{x}(y^2-x^2)}\,dx + \frac{\cancel{y^2-x^2}}{y\cancel{(y^2-x^2)}} + \frac{y^{\cancel{2}}}{\cancel{y}(y^2-x^2)}\,dy = 0$$

$$\left(\frac{1}{x} - \frac{x}{y^2-x^2}\right)dx + \frac{1}{y} + \frac{y}{y^2-x^2}\,dy = 0$$

above e.q. is exact D.E.

Now, find D.E

$$\int \frac{1}{x} - \frac{x}{y^2-x^2}\,dx + \int \frac{1}{y}\,dy = c$$

$$\int \frac{1}{x}\,dx - \int\left(\frac{x}{y^2-x^2}\right)dx + \int \frac{1}{y}\,dy = c$$

$$\log x + \frac{1}{2}\log(y^2+x^2) + \log y = c$$

$$\log(xy) + \log\sqrt{y^2-x^2} = c$$

$$\log\left(xy\sqrt{y^2-x^2}\right) = c$$

$$xy\sqrt{y^2-x^2} = c' \quad \text{.... Squaring.}$$

$$x^2y^2(y^2-x^2) = c''$$

$$\boxed{x^2y^2 - x^4y^2 = c_1}$$

Q13) Solve $x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$

→ Given $x^2 \frac{dy}{dx} = 3x^2 - 2xy + 1$

$$\cancel{\Rightarrow \frac{dy}{dx} = 3 - \frac{2y}{x} + \frac{1}{x^2}}$$

$$\frac{dy}{dx} = (3x^2 - 2xy + 1)\frac{1}{x^2}$$

$$\frac{dy}{dx} = \frac{3x^2}{x^2} - \frac{2xy}{x^2} + \frac{1}{x^2}$$

$$\frac{dy}{dx} = 3 - \frac{2y}{x} + \frac{1}{x^2}$$

$$\frac{dy}{dx} + \left(\frac{2}{x}\right)y = 3 + \frac{1}{x^2}$$

Above D.E is linear in y

Here $P = \frac{2}{x}$ $\quad Q = 3 + \frac{1}{x^2}$

$$I.f = e^{\int \frac{2}{x} dx}$$

$$I.f = e^{2\int \frac{1}{x} dx}$$

$$= e^{2\log x}$$

$$= e^{\log x^2}$$

$$I.f = x^2$$

G.S is given by

$$y \cdot e^{\int P \cdot dx} = \int Q \cdot e^{\int P \cdot dx} \, dx + C$$

$$y \cdot I.F = \int Q \cdot I.F. \, dx + C$$

$$y x^2 = \int \left(3 + \frac{1}{x^2}\right) x^2 \, dx + C$$

$$y x^2 = \int 3x^2 + \frac{x^2}{x^2} \, dx + C$$

$$x^2 y = \int 3x^2 + 1 \, dx + C$$

$$x^2 y = 3 \int x^2 dx + \int 1 \cdot dx + C$$

$$x^2 y = \cancel{3} \frac{x^3}{\cancel{3}} + x + C$$

$$\boxed{x^2 y = x^3 + x + C}$$

$$y = \frac{x^3 + x + C}{x^2}$$

$$y = \frac{x^{\cancel{3}}}{\cancel{x^2}} + \frac{x}{x^2} + \frac{C}{x^2}$$

$$\boxed{y = x + \frac{1}{x} + \frac{C}{x^2}}$$

14) Solve $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$

Solⁿ :→ $(x^2+1)\frac{dy}{dx} + 4xy = \frac{1}{(x^2+1)^2}$

$$\frac{dy}{dx} = \left(\frac{1}{(x^2+1)^2} - 4xy\right)\frac{1}{x^2+1}$$

$$\frac{dy}{dx} = \frac{1}{(x^2+1)^2(x^2+1)} - \frac{4xy}{x^2+1}$$

$$\frac{dy}{dx} + \left(\frac{4x}{x^2+1}\right)y = \frac{1}{(x^2+1)^3}$$

above eqⁿ is linear eqⁿ in y

$$P = \frac{4x}{x^2+1}, \quad Q = \frac{1}{(x^2+1)^3}$$

$$I.f = e^{\int \frac{4x}{x^2+1}dx}$$

$$I.f = e^{2\int \frac{2x}{x^2+1}dx}$$

$$I.f = e^{2\log(x^2+1)dx}$$

$$I.f = e^{\log(x^2+1)^2}$$

$$\boxed{I.f = (x^2+1)^2}$$

G.S is given by

$$y(I.f) = \int Q(I.f)\,dx + C$$

$$y(x^2+1)^2 = \int\left(\frac{1}{(x^2+1)^3}\right)(x^2+1)^2\,dx + C$$

$$y(x^2+1)^2 = \frac{1}{(x^2+1)}\,dx + C$$

$$\boxed{y(x^2+1)^2 = \tan^{-1}(x) + C}$$

Q15) Solve $\frac{dy}{dx} = \frac{e^x - 3xy}{x^2}$

→ Given $\frac{dy}{dx} = \frac{e^x - 3xy}{x^2}$

$$\frac{dy}{dx} = \frac{e^x}{x^2} - \frac{3xy}{x^2}$$

$$\frac{dy}{dx} + \left(\frac{3}{x}\right)y = \frac{e^x}{x^2} \quad \text{.....} \quad (1)$$

$$\frac{dy}{dx} + Py = Q \quad \text{where}$$

$$P = \frac{3}{x}, \quad Q = \frac{e^x}{x^2}$$

Now,

$$I.f = e^{\int f(x)dx}$$

$$I.f = e^{\int \frac{3}{x} dx}$$

$$I.f = e^{3\int \frac{1}{x} dx}$$

$$I.f = e^{3\log x}$$

$$I.f = e^{\log x^3}$$

$$I.f = x^3$$

Sol^n for first order linear D.E.

$$y.I = \int I.Q\, dx + C$$

$$y.x^3 = \int x^3 \frac{e^x}{x^2} dx + C$$

$$y.x^3 = \int x.e^x dx + C$$

$$y \cdot x^3 = \left[x e^x - \int 1 \cdot e^x \right] + c$$

$$y \cdot x^3 = x e^x - e^x + c$$

$$\boxed{y \cdot x^3 = (x-1) e^x + c}$$

Q16) $x \cos x \dfrac{dy}{dx} + (\cos x - x \sin x) y = 1$

→ Given $x \cos x \dfrac{dy}{dx} + (\cos x - x \sin x) y = 1$

Divide by $x \cos x$

$$\frac{dy}{dx} + \left(\frac{1}{x} - \tan x \right) y = \frac{1}{x \cos x} \quad \text{-----} \quad (1)$$

This is first order linear D.E of the form

$\dfrac{dy}{dx} + Py = Q$ where

$P(x) = \dfrac{1}{x} - \tan x \qquad Q(x) = \dfrac{\sec x}{x}$

$$I.f = e^{\int P(x) dx} = e^{\int (\frac{1}{x} - \tan x) \cdot dx}$$

$$If = e^{\log x - \log \sec x}$$

$$I.f = e^{\log \left(\frac{x}{\sec x} \right)}$$

$$= \boxed{\frac{x}{\sec x} = I}$$

Sol^n for first order linear D.E

$$y.I = \int I \, Q \, dx + c$$

$$\Rightarrow y \cdot \frac{x}{\sec x} = \int \left(\frac{x}{\sec x} - \left(\frac{\sec x}{x} \right) \right) dx + c$$

$$\Rightarrow y \cdot \frac{x}{\sec x} = x + c$$

$$\boxed{y x \cos x = x + c}$$

Q17) Solve $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

→ Given $xy - \frac{dy}{dx} = y^3 e^{-x^2}$

$$\frac{dy}{dx} - xy = -y^3 e^{-x^2} \quad \text{—— (I)}$$

where, $P = -x$, $Q = -e^{-x^2}$

Divide eqⁿ (I) by y^3

$$y^{-3}\frac{dy}{dx} - xy^{-2} = -e^{-x^2} \quad \text{—— (II)}$$

Put $y^{1-n} = u$ i.e. $\boxed{y^{-2} = u}$

diff w.r.t x.

$$\boxed{y^{-2} \longrightarrow y \longrightarrow x}$$

$$-2y^{-3}\frac{dy}{dx} = \frac{du}{dx}$$

$$y^{-3}\frac{dy}{dx} = -\frac{1}{2}\frac{du}{dx}$$

eqⁿ (II) becomes

$$-\frac{1}{2}\frac{du}{dx} - xu = -e^{-x^2}$$

$$\frac{du}{dx} + 2xu = 2e^{-x^2} \text{ ..... (multiply by } -2)$$

This is first order linear D.E

$$\frac{du}{dx} + Pu = Q \quad \text{with}$$

$$P = 2x \qquad Q = 2e^{-x^2}$$

$$IF = I = e^{\int P\,dx}$$
$$= e^{\int 2x\,dx}$$
$$= e^{2\left[\frac{x^2}{2}\right]}$$
$$I.f. = e^{x^2}$$

Sol^n for first order Linear D.E

$$uI = \int I Q\,dx + C$$
$$u.I = \int e^{x^2}\left(2e^{-x^2}\right)dx + C$$
$$u.I = \int 2e^{(x^2-x^2)}\,dx + C$$
$$u.e^{x^2} = 2\int 1.dx + C$$
$$u.e^{x^2} = 2x + C$$

Back substitute $u = y - 2$

$$\boxed{y^{-2}e^{x^2} = 2x + C} \quad \text{OR} \quad \boxed{\frac{e^{x^2}}{y^2} = 2x + C}$$

Q18) Solve $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)\,e^x \sec y$

$$\frac{dy}{dx} - \frac{\sin y}{\cos y\,(1+x)} = (1+x)\,e^x \sec y$$

multiply by $\cos y$

$$\cos y \frac{dy}{dx} - \frac{\sin y}{(1+x)} = (1+x)e^x \quad \ldots\ldots \quad (\sec y \cos y = 1) \quad \ldots\ldots \text{I}$$

This D.E of the form $f'(y)\frac{dy}{dx} + f(y)P = Q$.

$f(y) = \sin y$

put $\boxed{\sin y = u}$

$\cos y \frac{dy}{dx} = \frac{du}{dx}$

Eqn (1) becomes

$$\frac{du}{dx} - \frac{u}{(1+x)} = (1+x)e^x$$

This is first order Linear D.E of the type

$P = \frac{1}{1+x}$, $Q = (1+x)e^x$

$$I.f = e^{\int(-\frac{1}{1+x})}$$
$$= e^{-\log(1+x)}$$
$$e^{\log(1+x)^{-1}}$$

$$If = (1+x)^{-1}$$

$$\boxed{If = \frac{1}{(1+x)}}$$

Soln for first order Linear D.E

$$I.u = \int I\, Q\, dx + c$$

$$\frac{1}{1+x} \cdot u = \int I\,Q\,dx + C$$

$$\frac{1}{1+x} u = \int \cancel{(1+x)}\, e^x \frac{1}{\cancel{(1+x)}}\, dx$$

$$\frac{1}{1+x} \times \sin y = \int e^x$$

$$\sin y = (1+x)e^x + C$$

$$\boxed{\frac{\sin y}{1+x} = e^x + C}$$

Q19) Solve $\cos y - x\sin y \frac{dy}{dx} = \sec^2 x$

→ Given $\cos y - x\sin y \frac{dy}{dx} = \sec^2 x$

Divide by x

$$(-\sin y)\frac{dy}{dx} + \frac{1}{x}\cos y = \frac{\sec^2 x}{x} \quad \text{— (I)}$$

This D.E of the form $f(y)\frac{dy}{dx} + f(y)P = Q$

where P & Q are the functions of x (or) const

Here $f(y) = \cos y$ Put $\cos y = u$

$$\boxed{\cos y \to y \to u}$$

$$-\sin y \frac{dy}{dx} = \frac{du}{dx}$$

Eq^n (I) becomes

$$\frac{du}{dx} + \frac{1}{x}u = \frac{\sec^2 x}{x}$$

This is first order Linear D.E of the type $\frac{du}{dx} + Pu = Q$

where, $P = \frac{1}{x}$, $Q = \frac{\sec^2 x}{x}$

$$I.f = e^{\int P\,dx}$$
$$= e^{\int \frac{1}{x} dx}$$
$$= e^{\log x}$$
$$I.f = x$$

Sol^n for first order LDE

$$u.I = \int I.Q\,dx + C$$

$$ux = \int x \frac{\sec^2 x}{x}\,dx + C$$

$$ux = \tan x + C$$

Back Substitute $u = \cos y$

$$\boxed{\cos y\, u = \tan x + C}$$

Q20) Solve $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$

→ Given $3y^2 \frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$

Divide by $3y^2$

$$\frac{dy}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}\,y^{-2}}{3}$$

This is Bernoulli's D.E of the type

$$\frac{dy}{dx} + Py = Qy^n \quad \text{with } n=-2$$

where $P = \frac{2x}{3}$ $\quad Q = \frac{4xe^{-x^2}}{3}$

Put $y^{1-n} = u$

$y^3 = u$

diff w.r.t x

$$3y^3 \frac{dy}{dx} = \frac{du}{dx} \Rightarrow \frac{dy}{dx} = \frac{y^{-2}}{3}\frac{du}{dx}$$

$$\frac{dy}{dx} = \frac{1}{3y^2}\frac{du}{dx}$$

$$\frac{y^{-2}}{3}\frac{du}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}}{3}$$

$$\frac{1}{3y^2}\frac{du}{dx} + \left(\frac{2x}{3}\right)y = \frac{4xe^{-x^2}}{3y^2}$$

multiply by $3y^2$

$$\frac{du}{dx} + (2x)y^2 = 4xe^{-x^2}$$

but $y^3 = u$

$$\frac{du}{dx} + (2x)u = 4xe^{-x^2}$$

here, $P = 2x$, $Q = 4xe^{-x^2}$

$$IF = I = e^{\int P\,dx} = e^{\int 2x\,dx} = e^{2\left[\frac{x^2}{2}\right]} = e^{x^2}$$

Sol^n for first order Linear D.E

$$uI = \int IQ\,dx + C$$

$$ue^{x^2} = \int e^{x^2}(4xe^{-x^2})\,dx + C$$

i.e. $ue^{x^2} = 4\left[\frac{x^2}{2}\right] + C$

$ue^{x^2} = 2x^2 + C$

back substitute $u = y^3$

$$\boxed{y^3e^{x^2} = 2x^2 + C}$$